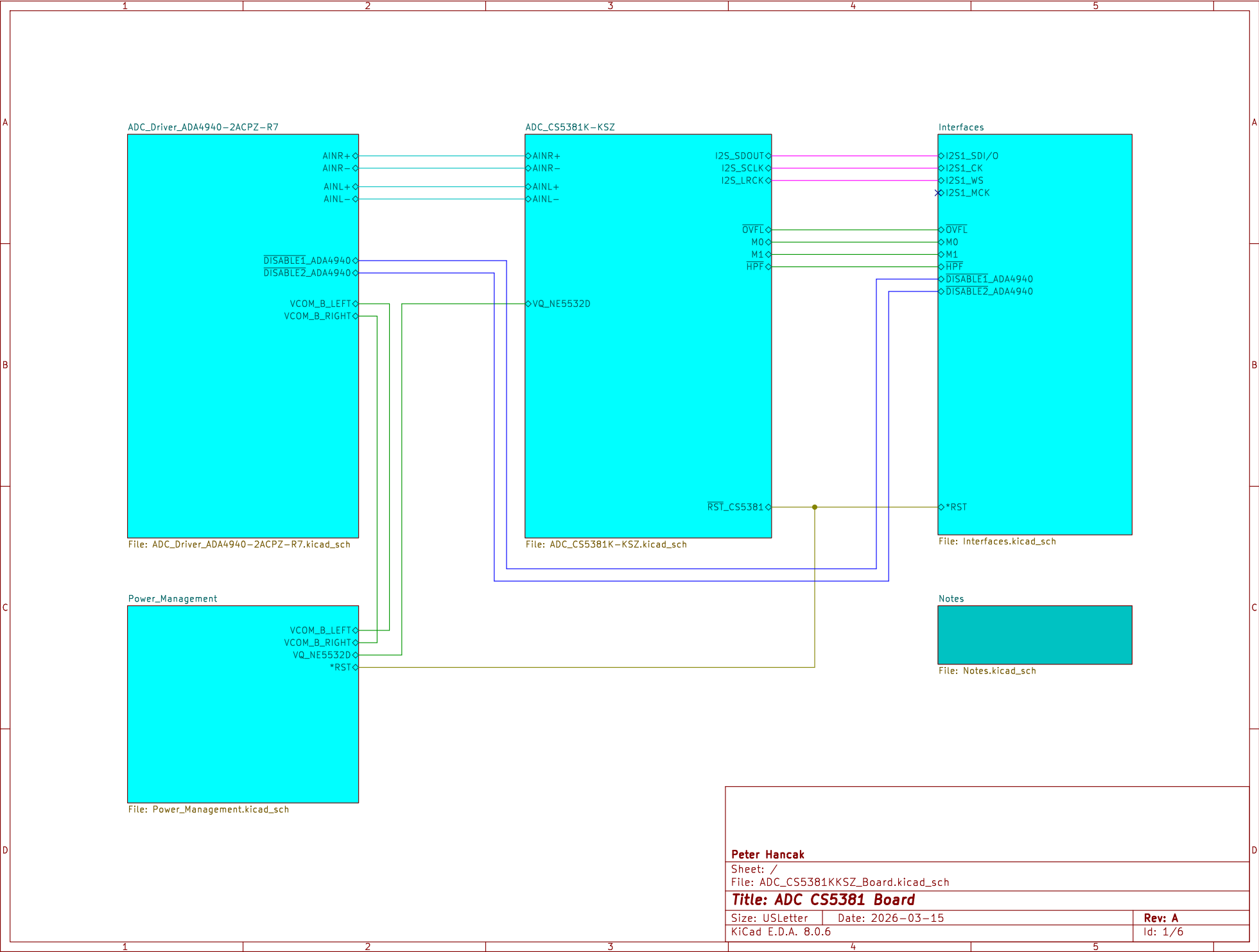
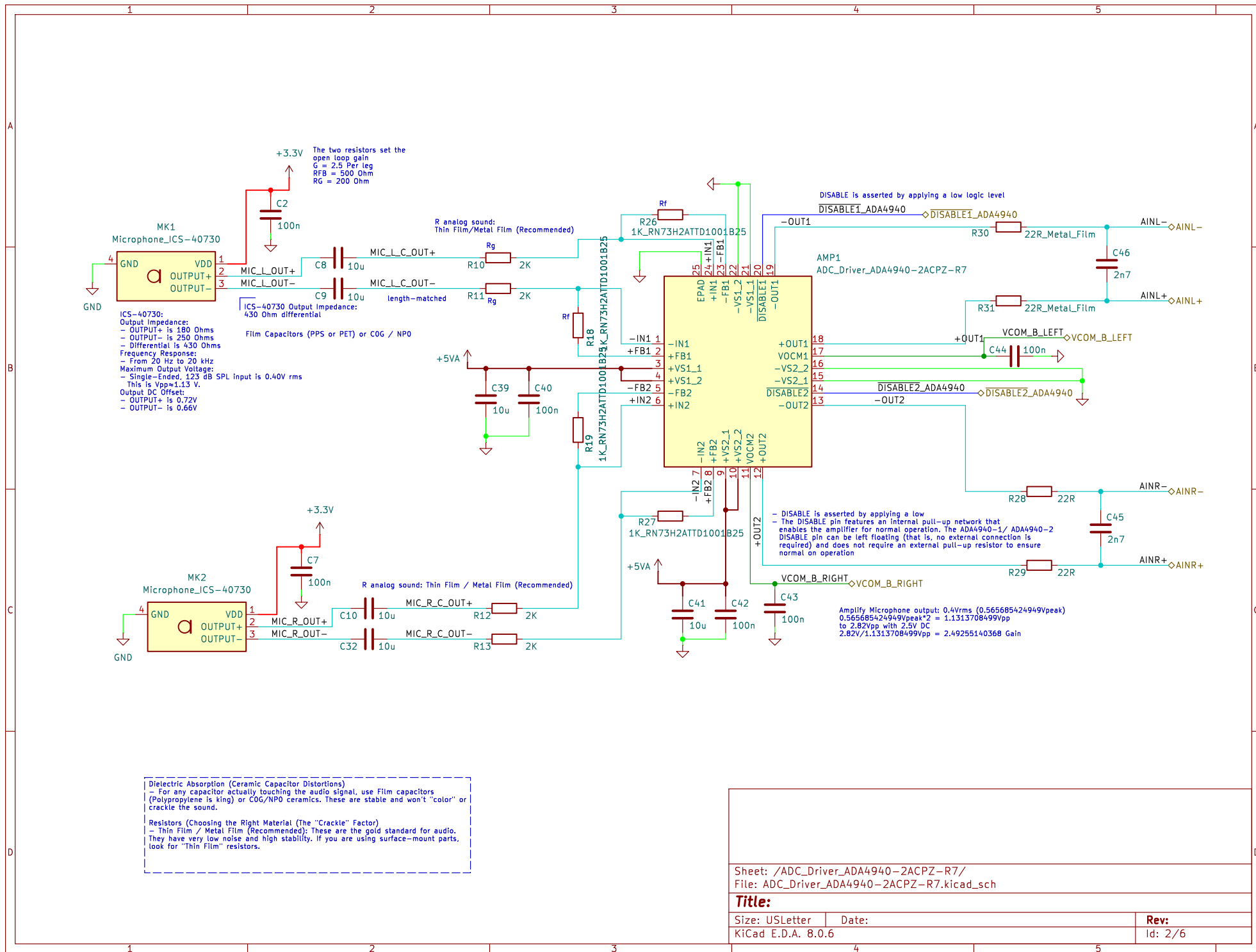




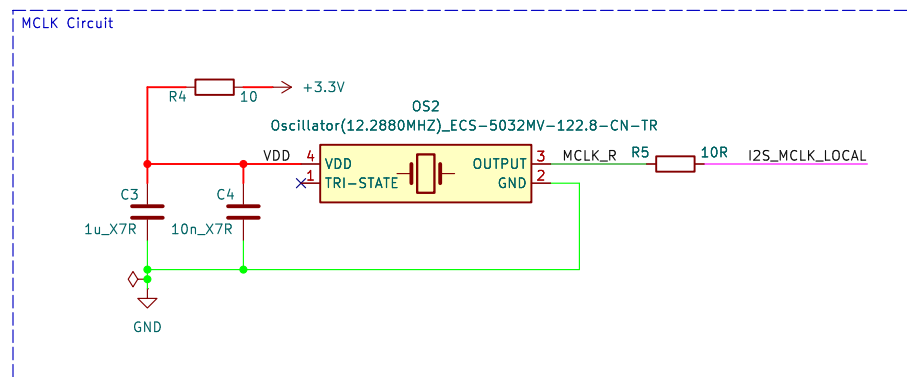
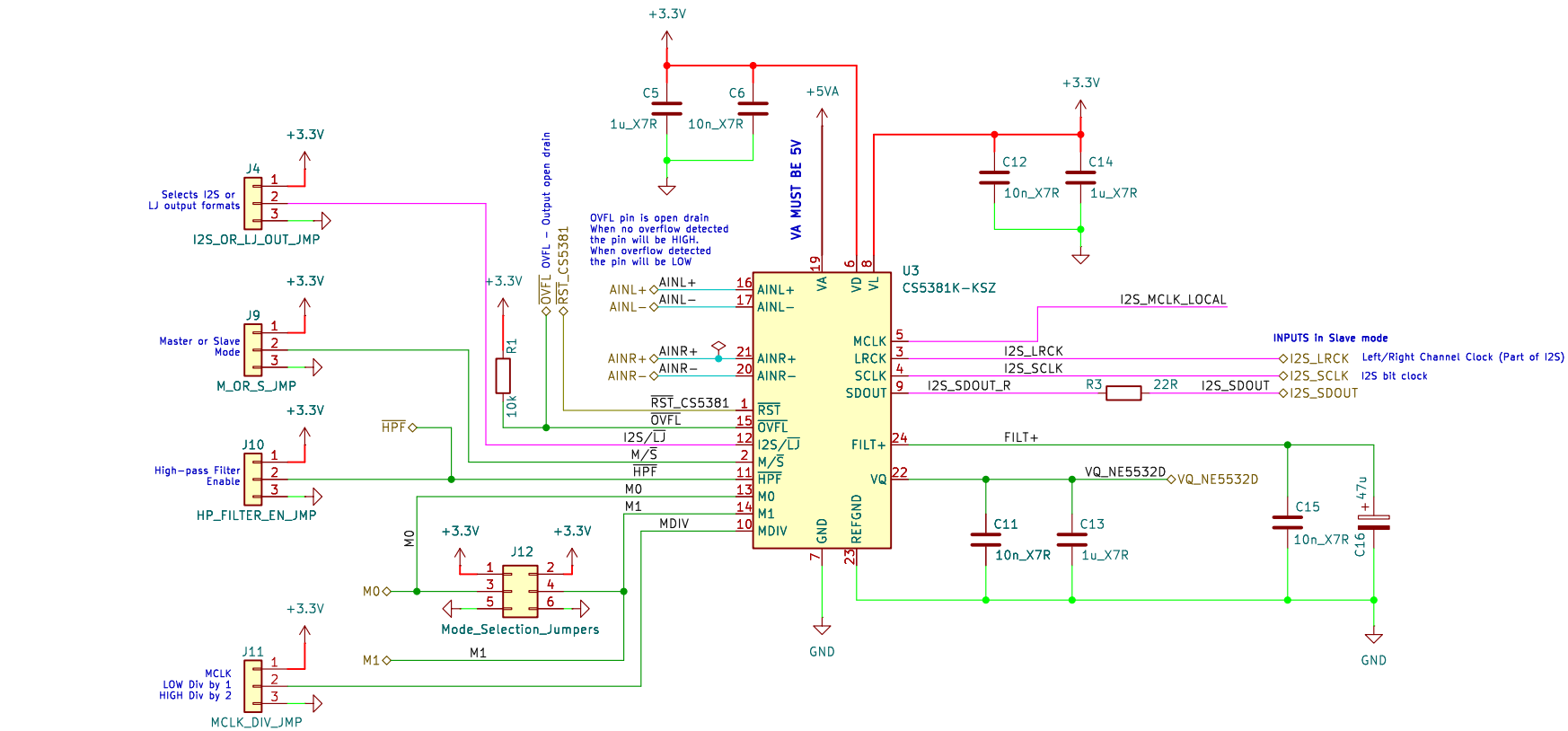


Sheet: /ADC_Driver_ADA4940-2ACPZ-R7/
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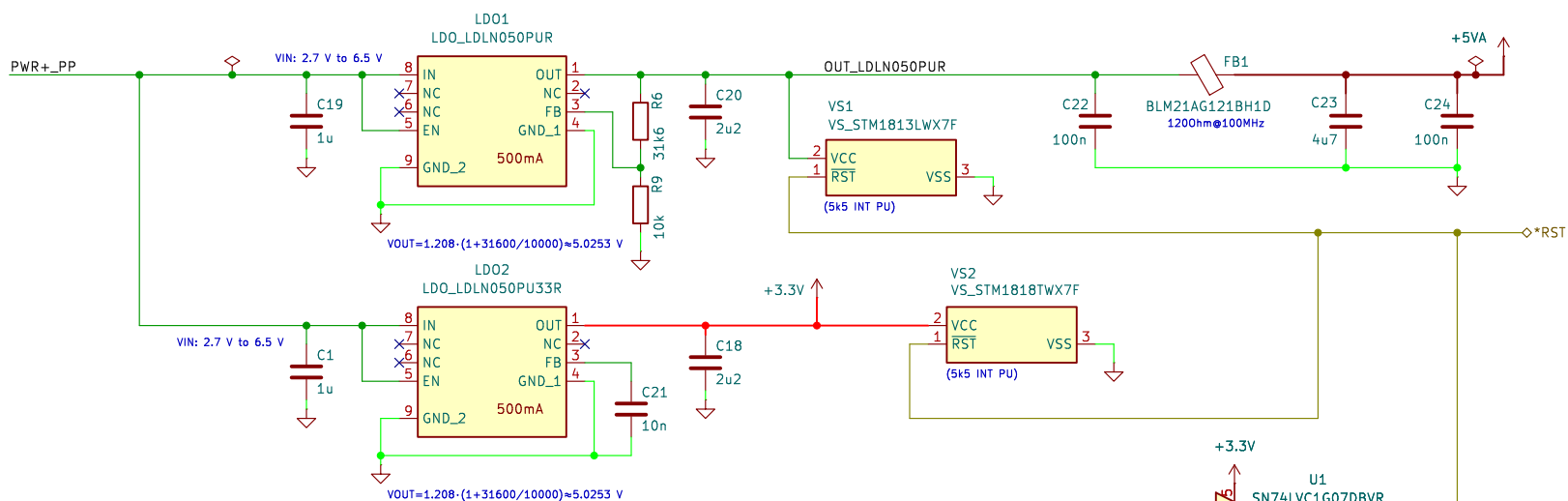
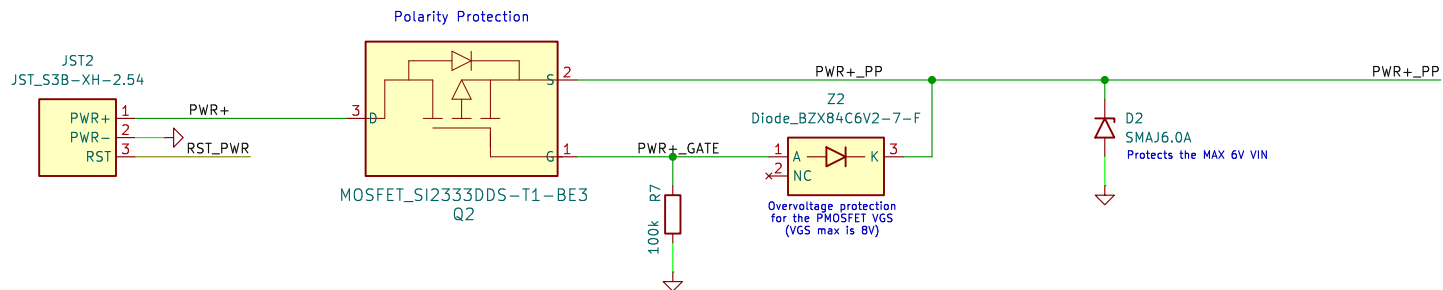
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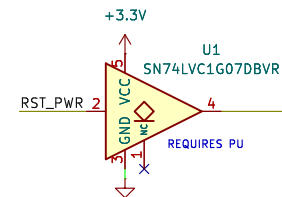
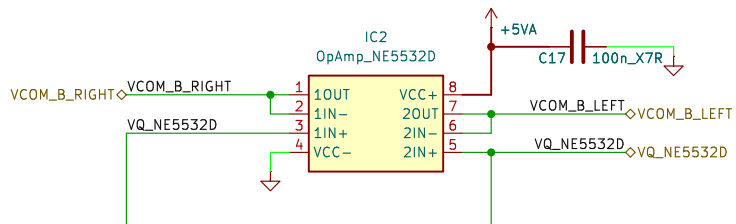
Notes:
The PG pins from both LDOs are connected to the RST pin.
When LDO voltage out are at the right levels the PG pins go High Impedance. While the level is lower than the desired Vout the PG pins are pulled to GND.
For CS4398 the RST pin should be pulled low at the start of the start up sequence.
If either LDO voltage level is not ready the RST pin will remain LOW.

Sheet: /ADC_CS5381K-KSZ/		File: ADC_CS5381K-KSZ.kicad_sch
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ON/OFF Switch, Polarity Protection, TVS



ADC Driver Input



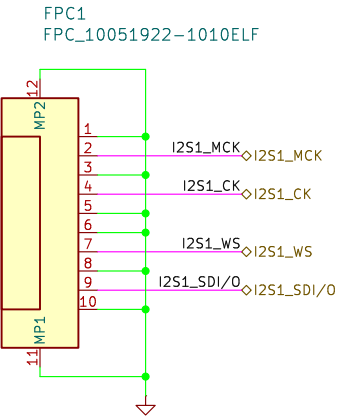
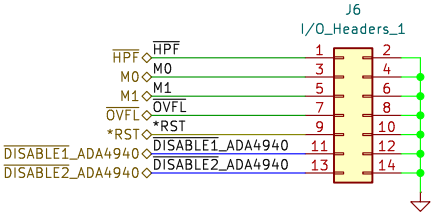
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Title:

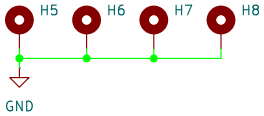
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Mounting Holes



Sheet: /Interfaces/ File: Interfaces.kicad_sch		
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Notes:

Physical power inputs: 3V3, 5V

Power Consumption

CS5381K-KSZ

IA = 50mA, ID = 50mA, ID = 30mA (MAX), ITOTAL = 130mA

ADC_Driver_ADA4940-2ACPZ-R7

I = 2.5mA

Microphones

I = 1mA

Device Total 150mA

All power inputs include polarity protection

Will be using 12.288 MHz to cover all possible conversion rates of CS5381

VA Analog MIN is 4.25V to 5.25V (Should be 5V)

VD Digital MIN is 3.1V to 5.25V

VL Logic MIN is 237V to 5.25V

VD pin powers the chip's digital core logic – the internal digital processing circuitry that handles things like: (Delta-sigma modulator output, Digital filters (decimation, etc.), Sample rate control logic, Configuration control logic (I²C/SPI core))

VL (Logic 1 = VL) controls MCLK, SCLK, LRCK

Power from USB: pins D+/D- left optionally floating or shorted

Fullscale Input Level = (Ain+)-(Ain-) = 5.64 Vpp

Ain+ is between 3.91 and 1.09 V (Each leg swinging 2.82V)

Ain- is between 3.91 and 1.09 V

Both oscillate around 2.5V (Common mode level)

The DC bias (Common mode level) is sett using VQ or VCOM (one pin)

VQ or VCOM nominal voltage is 2.5V

Filt+ Nominal Voltage 5V (Positive voltage reference output)

To reach 2.82 Vpp differential at the ADC:

Required gain

G≈2.82/2.26≈1.25 per leg (use two identical amps).

Differential Vpp scales by the same per-leg gain if both paths are equal.

If your target were 2.82 Vpp per input (single-ended):

G≈2.82/1.13≈2.5 per leg.

Impedance Matching between Microphone and ADA4940

Output Impedance to ICS-40730 is 430 Ohms

Input Differential Impedance to ADA4940-2 is Rin = 2 x 200 Ohms = 400 Ohms

The input impedance (dif) of ADA4940 should be 10x of souurce Impedance

10x 430 = 4k3 Ohms therefore make each Rg=2k

Design Factors

1. Power Supply

5V (Exact) through a connector

5V Through USB connector

LDO to create 3V3 for digital domain

2. Noise in the power supply (especially for CS5381 ADC)

Ferrite beads and capacitors

Maybe low noise LDO for the 3V3

3. ON/OFF Switch

On the 5V power input

4. ADA4940 to drive the CS5381 ADC

Amplify output from microphone ICS-40730 by G2.5 for CS5381 full range

5. Longer traces from microphone ICS-40730 to ADA4940 (about 5cm)

Impedance matching not needed

Length match

6. ADA4940 close to CS5381

Impedance matching not needed

7. Power input polarity protection

Using diode and mosfet

8. RST Trigger (Low power mode when LOW)

Local RST button

External pin header input

9. ADA4940 DISABLE PINS

Active LOW

The DISABLE pin features an internal pull-up network that enables the amplifier for normal operation

DISABLE pin can be left floating (that is, no external connection is required) and does not require an external pull-up resistor to ensure normal on operation

10. CS5381 ADC sampling at maximum 216kHz

MCLK is 216,000×24×2=10.368 MHz

Will transmit data through FPC ribbon cable

Fullscale Input Level = (Ain+)-(Ain-) = 5.64 Vpp

Ain+ is between 3.91 and 1.09 V (Each leg swinging 2.82V)

Ain- is between 3.91 and 1.09 V

Both oscillate around 2.5V (Common mode level)

The DC bias (Common mode level) is sett using VQ or VCOM (one pin)

VQ or VCOM nominal voltage is 2.5V

Filt+ Nominal Voltage 5V (Positive voltage reference output)

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